

FabLab / Innovation Center Shahpur University Planning

FabLab / Innovation Center — Design, Equipment & Network-Membership Proposal

For: A new university at Shahpur, India **Prepared by:** Vishvajeet N (drawing on hands-on experience at eirLab, Talence)
Reference models: eirLab (ENSEIRB-MATMECA, Talence/Bordeaux) · CFI — Center For Innovation (IIT Madras) · MIT / Fab Foundation FabLab standard **Date:** June 2026 · **Currency:** INR (USD reference where useful) **Scope of costing:** equipment + operating costs only. **Building / civil / fit-out costs are deliberately excluded** (assumed provided by the university as space + utilities).

One-line vision: *A student-run, open-access workshop where any student — regardless of branch or year — can walk in, learn to use a tool safely, and build a physical thing: a circuit, a robot, a 3D print, a laser-cut part, a product prototype.*

1. Guiding principles

These are the things that make CFI and eirLab work, distilled.

- Open access, not a locked teaching lab.** Long opening hours (ideally near-24/7 with student staff), no requirement to be enrolled in a specific course. This is the single biggest differentiator from a normal department lab — and, as Section 8 shows, it's also a hard requirement for joining the Fab Lab network.
- Student-run, faculty-backed.** A small core student team runs day-to-day operations (inventory, inductions, machine maintenance). One or two faculty mentors + one full-time technical staff member provide continuity. CFI is almost entirely student-governed; that ownership is *why* it thrives.
- Safety as the gate, not bureaucracy as the gate.** Access to a machine = pass a short safety induction for that machine. Everything else stays low-friction.
- Tiered tooling.** Cheap, abundant, beginner-safe tools (FDM printers, soldering, hand tools) in the open. Expensive/dangerous tools (laser, CNC, table saw) behind induction + supervision.
- Start lean, design for growth.** Phase the equipment buy. A working lab students love beats an empty showpiece.
- Project culture over inventory.** Budget for competitions (Robocon/ABU, Formula Student, e-Yantra, hackathons), club projects, and consumables — not just one-time machines.

2. Reference models — what to borrow from each

Model	What it is	Borrow this
eirLab (Talence)	University FabLab at ENSEIRB-MATMECA, part of the global Fab Foundation network	Fab Foundation equipment standard, induction/badge system, open membership, digital fabrication focus
CFI (IIT Madras)	Large student-run innovation center with themed clubs (Aero, Robotics, 3D printing, Electronics, etc.)	Student governance, club structure, competition teams, 24/7 access culture
MIT / Fab Foundation	The original "Fab Lab" inventory spec + global network	A globally recognised, vendor-neutral equipment checklist; membership in the international network (Section 8)

Recommendation: Adopt the **Fab Foundation equipment baseline** (so you can call it a real "FabLab" and join the network), but organise the *people* and *culture* the **CFI way** (student clubs + competition teams + open access).

3. Space requirements (university to provide)

Space and civil works are assumed to be provided by the university; no building cost is included here. What the lab *needs* from the space:

3.1 Size

Tier	Usable area	Concurrent users
Student	200 m ² / 2,150 ft ²	20-40

Starter	~200 m ² (~2,150 ft ²)	30–40
Standard FabLab (recommended target)	~500 m ² (~5,400 ft ²)	60–90
Flagship (CFI-scale)	1,000–1,500 m ²	150+

A single large hall plus 2–3 partitionable rooms is enough to start. Ground-floor access (for moving heavy machines/stock) is strongly preferred.

3.2 Zoning (noise, dust, fumes, and safety segregation)

ENTRANCE / RECEPTION + SAFETY INDUCTION DESK + LOCKERS		
DRY ZONE (quiet) CAD machines, meeting/ teaching nook	CLEAN DIGITAL FAB (3D print farm, laser cutter w/ extraction, vinyl cutter)	ELECTRONICS & ROBOTICS LAB benches, soldering scopes, PSUs, component store
WET / DIRTY ZONE (separate ventilation, dust extraction) CNC router · woodworking · metal/bench tools · welding bay · spray/paint corner · grinder		
STORAGE: filament/material store · tool crib · finished project shelving · hazmat cabinet		

3.3 Utilities the space must supply (no cost charged to lab budget here)

- **3-phase power** + distributed sockets, RCD/ELCB protection, separate circuits for high-draw machines (laser, CNC, welder).
- **Fume & dust extraction routing** (the extraction *equipment* is in the lab budget, Section 4.5; the ducting/openings to outdoors are a building item).
- **Ventilation & AC** for the dry/electronics/digital zones; the wet zone can be naturally ventilated.
- **Compressed air** point for the dirty zone.
- **Fire-safety provisioning**, marked exits, water point for eyewash.
- **ESD-safe flooring** in the electronics zone; epoxy/concrete in the dirty zone.
- **Wide door / roll-up shutter** to the dirty zone.

4. Equipment list — best-in-class spec (India 2026 estimates)

Procurement philosophy (the "HPC mentality"): buy *professional-grade, serviceable, safety-first* machines, not the cheapest unit that technically works. Where a marginal ("epsilon") spend buys real gains in **uptime, output quality, repairability, local service/warranty, or operator safety**, we take the upgrade — because in an open, high-throughput student lab the lifetime cost of a flaky machine (downtime, scrapped work, frustrated users, safety incidents) dwarfs the purchase price. Named models below are concrete *anchors* for quality/price; equivalent-tier alternatives are listed so procurement can competitively quote. This list also satisfies the Fab Foundation common-inventory requirement (§8) — several picks (ShopBot, Roland mill/vinyl) *are* the Fab standard machines.

4.1 Digital fabrication (the FabLab core)

Equipment	Qty	Unit (₹)	Subtotal (₹)	Best-in-class anchor & why
FDM 3D printer — enclosed CoreXY + multi-material	4	1,30,000	5,20,000	Bambu Lab X1-Carbon + AMS — fastest reliable prosumer, multi-colour/material, hardened nozzle for CF. The student workhorse.
FDM 3D printer — open, ultra-serviceable	2	1,00,000	2,00,000	Prusa MK4S — open-source, every part replaceable in-house, runs for years. Insurance against vendor lock-in.
Engineering FDM — dual-extrusion, soluble support	1	7,50,000	7,50,000	UltiMaker S7 Pro bundle — heated chamber, PVA support, true engineering polymers (PC, PA, PETG-CF).
Composite FDM — continuous carbon/glass fibre	1	12,00,000	12,00,000	Markforged Mark Two / Onyx Pro — prints metal-strength functional parts (jigs, drone frames, robotics arms). Nothing else does this in-house. (<i>Flagship item — high value-per-rupee for serious teams.</i>)
Resin SLA — precision + closed ecosystem	1	5,00,000	5,00,000	Formlabs Form 4 + Form Wash + Form Cure — best surface finish & repeatability, engineering/dental/castable resins.
Large MSLA resin +	1	70,000	70,000	Elegoo Saturn 4 Ultra (+station) — high-volume detailed prints at low cost.

wash/cure				
CO ₂ laser cutter 100 W + chiller + premium extraction	1	12,00,000	12,00,000	Thunder Laser Nova 51 (RECI tube) — metal frame, autofocus, rock-solid; pairs with BOFA extraction (\$4.5). (<i>Trotec Speedy is the ~₹30 L step-up.</i>)
Fibre laser marker/cutter (metal)	1	4,50,000	4,50,000	Raycus/JPT 30-50 W fibre — engrave & mark metal, cut thin sheet; opens metalwork the CO ₂ can't.
CNC router — full-sheet (the Fab standard)	1	18,00,000	18,00,000	ShopBot PRSalpha 96×48 — the canonical Fab Lab CNC; furniture, large enclosures, mould-making.
Desktop precision mill — PCBs + small metal	1	4,50,000	4,50,000	Roland SRM-20 (Fab standard) — clean single-/double-sided PCBs, wax/aluminium machining.
Desktop PCB prototyping mill (fast turn)	1	5,50,000	5,50,000	Bantam Tools Desktop PCB Mill — same-hour PCB iteration; huge for electronics/robotics teams.
Vinyl cutter (the Fab standard)	1	1,80,000	1,80,000	Roland CAMM-1 GS-24 — stickers, stencils, flexible copper, vinyl masks for etching.
Subtotal			78,70,000	

4.2 Electronics & robotics

Equipment	Qty	Unit (₹)	Subtotal (₹)	Best-in-class anchor & why
Soldering station — reference grade	4	55,000	2,20,000	JBC CD-2BHE — fastest tip recovery, best joints, near-indestructible. The bench everyone fights over.
Soldering station — quality workhorse	4	25,000	1,00,000	Hakko FX-951 — reliable, cheap tips, perfect for the rest of the benches.
Hot-air rework stations	2	60,000	1,20,000	Hakko FR-810B — SMD/BGA rework done properly.
Oscilloscope — reference (350 MHz, 4-ch MSO)	1	5,50,000	5,50,000	Keysight InfiniiVision DSOX3034T — the gold-standard teaching/debug scope.
Oscilloscope — high-value 4-ch MSO	3	1,80,000	5,40,000	Rigol MSO5074 — pro features at a fraction of the price; great for the general benches.
Triple-output bench PSU	4	80,000	3,20,000	Keysight E36312A — clean, programmable, protected.
Arbitrary function generator	2	1,80,000	3,60,000	Keysight 33500B — precise signals for real characterisation.
Bench DMM — 6½ digit	2	1,80,000	3,60,000	Keysight 34465A — lab-reference measurements.
Handheld DMM	6	30,000	1,80,000	Fluke 87V — the industry-standard rugged meter.
Logic analyser	2	1,00,000	2,00,000	Saleae Logic Pro 16 — best protocol-decode UX, debugs any digital bus.
Spectrum analyser	1	2,80,000	2,80,000	Siglent SSA3021X — RF/EMC debugging for wireless & power projects.
SMD assembly line — P&P + reflow + stencil printer	1 set	9,50,000	9,50,000	Neoden 4 pick-and-place + reflow oven + stencil printer — in-house small-batch PCB assembly.
MCU / SBC / FPGA pool	pool	—	5,00,000	NVIDIA Jetson AGX Orin + several Orin Nano, RPi 5, ESP32, STM32, Teensy 4, Digilent FPGA boards — real edge-AI & embedded work.
Robotics platform pool	pool	—	9,00,000	Desktop robot arm (UFACTORY xArm 6 / Dobot CR), Dynamixel smart servos, Intel RealSense + Luxonis OAK-D vision, Livox/RPLidar , drone dev kits, LiPo packs + smart chargers. Feeds Robocon/FS/e-Yantra teams.
Component inventory (genuine parts, SMD reels) + ESD storage	—	—	3,00,000	Quality, traceable parts; treat partly as consumable (\$6).
Subtotal			59,00,000	

4.3 Workshop / dirty zone

Equipment	Qty	Unit (₹)	Subtotal (₹)	Best-in-class anchor & why
Table saw — flesh-detection safety	1	4,50,000	4,50,000	SawStop — stops the blade in <5 ms on skin contact. In a student lab this single upgrade prevents life-changing injuries; non-negotiable.
Track saw + integrated dust extraction	1 set	1,60,000	1,60,000	Festool TS 55 + CT extractor — precise sheet breakdown, near-zero dust.
Bandsaw, drill press, mitre saw (quality)	set	—	3,20,000	Record Power / Bosch Professional class — accurate, durable, induction-gated.

Hand & power tools	set	—	3,00,000	Festool / Bosch Professional / Makita — tools that survive shared abuse and stay accurate.
Bench grinder, belt/disc sander	set	—	1,00,000	Quality, well-guarded units.
Multiprocess welder (MIG/TIG/stick) + auto-darkening helmets + welding fume extractor	1 set	3,80,000	3,80,000	Miller/Lincoln/ESAB multiprocess — proper welds for robotics/auto teams; dedicated fume capture.
Heavy workbenches + quality vises	4	40,000	1,60,000	Steel benches that take a beating.
Industrial sewing + embroidery + heat press	set	—	1,50,000	Soft fabrication, wearables, drone covers, composite layup.
Subtotal			20,20,000	

4.4 Computing, design & metrology

Equipment	Qty	Unit (₹)	Subtotal (₹)	Best-in-class anchor & why
CAD/CAM/sim workstations	6	3,00,000	18,00,000	Dell Precision / HP Z class, RTX 4080/A-series, 64-128 GB — heavy CAD, CAM, FEA/CFD, slicing without waiting.
GPU node for AI/sim training	1	4,50,000	4,50,000	RTX 4090 / dual-GPU — robotics vision, ML, large simulations.
Teaching display + 4K monitors	set	—	2,00,000	Large interactive display/projector + 4K screens for the teaching nook.
3D scanner — pro	1	6,50,000	6,50,000	EinScan HX (or Artec Leo at the ~₹18 L step-up) — accurate reverse-engineering & metrology-grade capture.
Metrology set	set	—	2,80,000	Mitutoyo calipers/micrometers/height gauge + measuring microscope — the trusted standard.
Subtotal (capital)			33,80,000	Software is OpEx (\$6); lean on FOSS (FreeCAD, KiCad, Inkscape) + a few pro seats (Fusion 360, SolidWorks education).

4.5 Safety & shared equipment

Item	Subtotal (₹)	Best-in-class anchor & why
Fume extraction — laser + multi-arm solder	6,50,000	BOFA filtration — the industry premium; protects lungs and keeps the lab open during inspections.
Dust extraction (cyclone + ducting + drops)	2,80,000	Proper HEPA cyclone system for CNC/woodworking.
PPE (quality, sized, replenished)	1,20,000	Goggles, gloves, aprons, ear/lung protection, face shields, welding gear.
Fire suppression, smoke/heat detectors, eyewash, first aid	1,80,000	Including a CO ₂ /clean-agent extinguisher near electronics & resin.
Signage, induction materials, machine SOP cards, labelling	60,000	
RFID access-control system	2,20,000	The eirLab-style "earn the badge" per-machine access model.
Subtotal	15,10,000	

5. Equipment cost summary (best-in-class build — equipment only)

Block	Cost (₹)
4.1 Digital fabrication	78,70,000
4.2 Electronics & robotics	59,00,000
4.3 Workshop / dirty zone	20,20,000
4.4 Computing & metrology	33,80,000
4.5 Safety & shared equipment	15,10,000
Equipment subtotal	2,06,80,000
Contingency (~10%)	~20,70,000
TOTAL equipment capital	≈ ₹2.27 crore (~US\$267k)

Three build levels (equipment only) — so the budget can be staged or trimmed:

Build	What you get	Equipment capital (₹)	USD approx
Lean / by-the-book Fab Lab	Fab Foundation core inventory, value-grade brands	35-55 lakh	\$41-65k

Best-in-class (recommended)	the spec above — pro-grade, safety-first, edge-AI & composite capable	₹2.0-2.3 cr	\$235-270k
— of which "flagship stretch" items	Markforged composite, fibre laser, ShopBot, Neoden P&P, Artec scanner	~50-60 lakh	\$59-71k

The five "flagship stretch" items deliver capabilities nothing cheaper can match; if cash-flow demands, defer them to Phase 2 and the build still lands at a strong **≈ ₹1.6-1.7 cr** while keeping every core machine at top quality. Building, civil, electrical, HVAC and furniture remain excluded (university-provided).

5a. Benchmarked against the official Fab Foundation equipment standard

The Fab Foundation publishes canonical "what does a Fab Lab cost" figures for **machines + consumables only** (source: [Getting Started with Fab Labs](#), referenced from [Growing the Fab Lab Network](#)). At ~₹85/US\$ (2026):

Fab Foundation official figure	USD	INR (≈ ₹85/\$)
Capital equipment (MIT inventory list)	\$25,000 – \$65,000	₹21 – 55 lakh
Initial consumables	\$15,000 – \$40,000	₹13 – 34 lakh
Subtotal to stand up a "regular" Fab Lab	\$40,000 – \$105,000	₹34 – 89 lakh
Mobile Fab Lab (reference only)	\$298,000	≈ ₹2.5 crore

(UK chapter quotes the equivalent £30-80k capital + £10-25k consumables.)

How this proposal maps onto the benchmark: the Fab Foundation range describes a *baseline* lab built with value-grade machines. This proposal deliberately spends **~3-4× that baseline (≈ ₹2.0-2.3 cr)** because we chose **professional-grade, serviceable, safety-first equipment** and added capabilities the bare inventory omits — continuous-fibre composite printing, a metal-capable fibre laser, in-house PCB assembly (pick-and-place), flesh-detection saw safety, edge-AI/robotics platforms, and metrology-grade scanning. The result is still a fully **Fab-Foundation-compliant** lab (it contains every standard machine — ShopBot, Roland mill/vinyl, etc.), just specced at the quality tier of a top university makerspace rather than the entry minimum. A lean, by-the-book version remains available at **₹35-55 lakh** (table in §5) if budget is constrained.

6. Operating cost (annual, best-in-class build)

Premium machines cost a little more to run (genuine consumables, AMC/service contracts) but break far less often — the trade is fewer catastrophic failures and far higher uptime.

Item	Annual (₹)
Full-time technical staff (1-2: lab engineer + technician)	8-14 lakh
Consumables (filament, resin, sheet stock, genuine SMD parts, blades, bits, PPE refills)	8-14 lakh
Maintenance, AMC & spares (laser tube, Formlabs/UltiMaker service, ShopBot bits, calibration)	5-10 lakh
Software (mostly FOSS; a few pro seats — Fusion 360, SolidWorks education)	1-3 lakh
Power (high-draw machines + GPU node)	4-7 lakh
Competitions, project grants, hackathons, travel	5-15 lakh
Fab network participation (membership free; Fab Academy tuition optional — see §8)	0-8 lakh
Total OpEx	≈ ₹31-71 lakh / year

Sustainability levers: small per-print/material charge-back at cost; sponsored club teams; industry-sponsored machines (named bays); paid weekend workshops for the public/schools (also satisfies the public-access requirement, §8); consultancy/prototyping for local startups (eirLab and many FabLabs do this).

7. Phasing plan (equipment)

Phase 0 — Pre-launch (Months 0-3) University allocates space + utilities; recruit 1 faculty mentor + 1 technical staff + a 6-8 student core team; draft safety SOPs and the induction/badge system; finalise Phase-1 procurement; **publicly endorse the Fab Charter** (§8).

Phase 1 — "Get students building" (Months 3-8) · equipment ≈ ₹70-90 lakh Commission the everyday core at full quality: FDM print farm (Bambu ×4 + Prusa ×2), Formlabs resin, the CO₂ laser (+BOFA extraction), premium electronics benches (JBC/Keysight/Rigol), workstations, SawStop + core power tools, the MCU/robotics pool, and the full safety kit.

Open the doors with public-access hours. This alone is a complete, lovable, top-tier makerspace and already meets the Fab common-tools criterion.

Phase 2 — "Full FabLab + flagship capabilities" (Months 8-18) • +equipment ≈ ₹1.2-1.4 cr Add the ShopBot CNC, Markforged composite printer, fibre laser, UltiMaker engineering FDM, Neoden pick-and-place line, Roland/Bantam mills + vinyl, EinScan scanner, welding bay, and GPU node. **Register on fablabs.io and join the Fab Foundation network** (§8). Launch competition teams (Robocon, e-Yantra, Formula Student feeder).

Phase 3 — "Scale & specialise (CFI-style)" (Year 2+) Welding/metal bay, dedicated club rooms, industry-sponsored bays, public/school outreach, startup incubation tie-in, optional Fab Academy node.

8. Joining the global Fab Lab network (Fab Foundation) — requirements & application procedure

Becoming a recognised Fab Lab is **free to register** — the "cost" is meeting the criteria below (the equipment in §4 and open-access hours in §1 already do most of the work). Membership gives you the right to use the **FabLab name and logo** (useful for fundraising), a pin on the **world map**, access to the **Fab Academy** programme, and participation in the global community.

8.1 The four requirements (from the Fab Charter)

A space qualifies as a Fab Lab when it meets **all four**:

1. **Public access.** The lab must be open to the public — free or on an in-kind/barter basis — **at least part of the time each week**. (Our open-access model in §1 satisfies this; formalise specific public hours.)
2. **Subscribe to the Fab Charter.** Operate by, and publicly endorse, the [Fab Charter](#) — its statements on mission, access, responsibilities, secrecy (designs/processes are shared), and business.
3. **Share the common tool set & processes.** Use broadly the same core machines and software as other Fab Labs (the MIT inventory) so that a project made in one lab can be reproduced in another. Our §4 list is built to this standard.
4. **Participate in the Fab Lab network.** Engage with the wider community — global video conferences, the annual **Fab** conference, knowledge-sharing, and (optionally) hosting/sending students to **Fab Academy**.

8.2 Step-by-step application procedure

1. **Build to the four criteria.** Confirm the §4 inventory covers the common tool set, and commit to weekly public-access hours.
2. **Endorse the Fab Charter** publicly (web page / signage in the lab).
3. **Create a personal account** on the network registry at fablabs.io.
4. **Register the lab** at fablabs.io/labs/new — add location, photos, equipment, contact, and opening hours. The lab appears on the [world map](#).
5. **Set status** **planned** → **active** once the lab is operational and meeting public-access hours. Submissions pass through a community/regional **approval** step.
6. **Connect with the regional network.** For India/South Asia, engage with the existing Indian Fab Lab community (e.g. **Vigyan Ashram, Pabal** — Asia's first Fab Lab) and the regional Fab coordination for endorsement and mentoring.
7. **Plug into Fab Academy (optional but recommended).** Enrol your technical staff / lead students in **Fab Academy** ("How To Make (Almost) Anything"), or apply to become a **local node** that hosts students. This is the strongest signal of network participation and trains your team on the standard toolchain. (*Budget note: Fab Academy tuition has historically been on the order of US\$5k/student/year; hosting as a node spreads this — reflected as the optional ₹0-8 lakh line in §6.*)
8. **Use the logo** in fundraising and signage once you meet all criteria and are "in synchrony with the Fab Lab form and spirit," as the Fab Foundation phrases it.

8.3 What it costs (network side)

Item	Cost
Registering on fablabs.io / world-map listing	Free
Using the FabLab name & logo (once compliant)	Free
Meeting the common-inventory requirement	Already in §4 equipment budget
Committing to public-access hours	Operational policy, no capital cost
Fab Academy (optional, per student or as a node)	~US\$5k/student/yr historically — optional

9. People & governance

- **Faculty mentor(s):** 1-2, light-touch, unblock funding/space, sign off on safety policy.
- **Lab engineer / manager (full-time):** keeps machines alive, owns safety, trains the student core.
- **Student core team (8-15):** elected/selected; portfolios = Operations, Inventory, Safety/Inductions, each major machine, Outreach, Finance. *This is the CFI secret sauce.*
- **Member students:** anyone who passes the general safety induction; per-machine badges unlock the gated tools.
- **Clubs/teams under the umbrella:** Robotics, Aero/Drones, 3D-printing, Electronics/IoT, Auto, Software/AI — each owning projects and competition entries.

Access model (eirLab-style): General induction → member badge → per-machine induction → machine badge → RFID access during open hours. Dangerous machines additionally require a trained student/staff "buddy" present.

10. Risks & how to avoid the common failure modes

Risk	Mitigation
Becomes a locked "showpiece" lab nobody uses	Open access + student governance from day one; measure footfall, not inventory (also a \$8 requirement)
Extraction/ventilation skipped → laser/solder fumes, dust	Keep extraction equipment in the lab budget (\$4.5); ensure the space provides ducting
Machines break and sit idle	AMC + spares budget + trained student maintainers; buy <i>fewer, repairable, well-supported</i> brands
No consumables budget → machines unusable	Ring-fence annual OpEx; introduce at-cost material charge-back
Faculty over-control kills student ownership	Mentor, don't manage; let the student core run operations
Safety incident shuts it all down	Induction gating, PPE, SOP cards, supervision on dangerous tools, insurance

11. One-paragraph executive summary (for the proposal cover)

We propose an **open-access, student-run FabLab & Innovation Center** at the new Shahpur campus, modelled on eirLab (Talence) for its digital-fabrication discipline and IIT Madras's CFI for its student-governed, competition-driven culture, and built to the **MIT/Fab Foundation common-inventory standard** so it can **join the global Fab Lab network**. Equipment is specced **best-in-class** — professional-grade, serviceable, safety-first — including continuous-fibre composite 3D printing, a full-sheet ShopBot CNC, CO₂ + fibre lasers, in-house PCB pick-and-place, flesh-detection saw safety, and edge-AI/robotics platforms. Using university-provided space, the **equipment outlay is ≈ ₹2.0-2.3 crore** (~US\$235–270k; phased, with a complete top-tier Phase-1 makerspace for ≈ ₹70–90 lakh; a lean by-the-book alternative is ₹35–55 lakh), delivering 3D printing, laser/CNC cutting, electronics, robotics and a workshop accessible to **every student** via a simple safety-induction badge system. **Annual operating cost ≈ ₹31-71 lakh**. Network registration on fablabs.io is **free** once the four Fab Charter criteria are met. The center becomes the campus's hands-on engine for tinkering, prototyping, competitions, and student startups — and a recognised node on the worldwide Fab Lab map.

Notes: All costs are 2026 indicative estimates for India (INR), exclusive/inclusive of GST as marked by vendors, and should be validated with 2–3 quotations per major machine before budgeting. USD conversions at ~₹85/\$. Building, civil, electrical, HVAC and furniture costs are intentionally excluded.

Sources: [Fab Foundation — Getting Started](#) · [The Fab Charter \(MIT CBA\)](#) · [fablabs.io registry](#) · [Fab Foundation — global community](#) · [Fab Foundation UK — how to set up a FabLab](#)